

# AUDIO OUTPUT

I2S DAC -> PAM8403 Class-D Amplifier -> 28mm 8ohm Speaker

## I2S Bus (PDM TX mode - DOUT only):

I2S\_BCLK

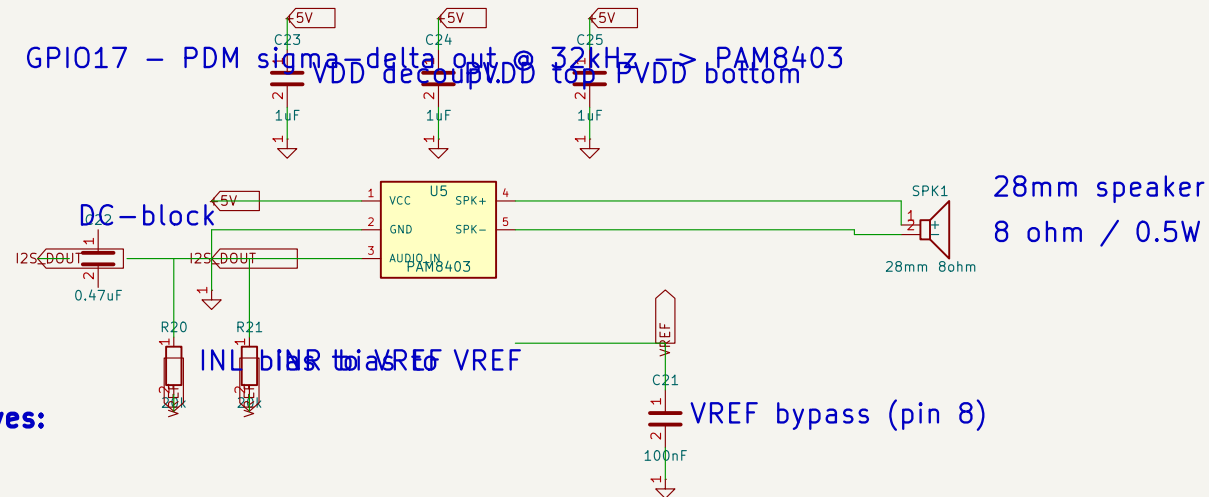
GPIO15 - Bit Clock (UNUSED - PDM mode, v2 free)

I2S\_LRCK

GPIO16 - L/R Word Select (UNUSED - PDM mode, v2 free)

I2S\_DOUT

GPIO17 - PDM sigma-delta out @ 32kHz -> PAM8403



## PAM8403 Passives:

## Design Notes:

- PAM8403: filterless Class-D stereo amplifier (3W per channel max)
- Only one channel used for mono audio
- No external DAC needed: ESP32-S3 has built-in I2S with DMA
- Powered from +5V rail for max headroom (3.3V limits volume)
- DMA-driven audio streaming for low CPU overhead

Generated by scripts/generate\_schematics  
ESP32 Emu Turbo - Handheld Retro Console

Sheet: /  
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